

CONSTRAINTS AND JOINS QUERIES:

task 1:

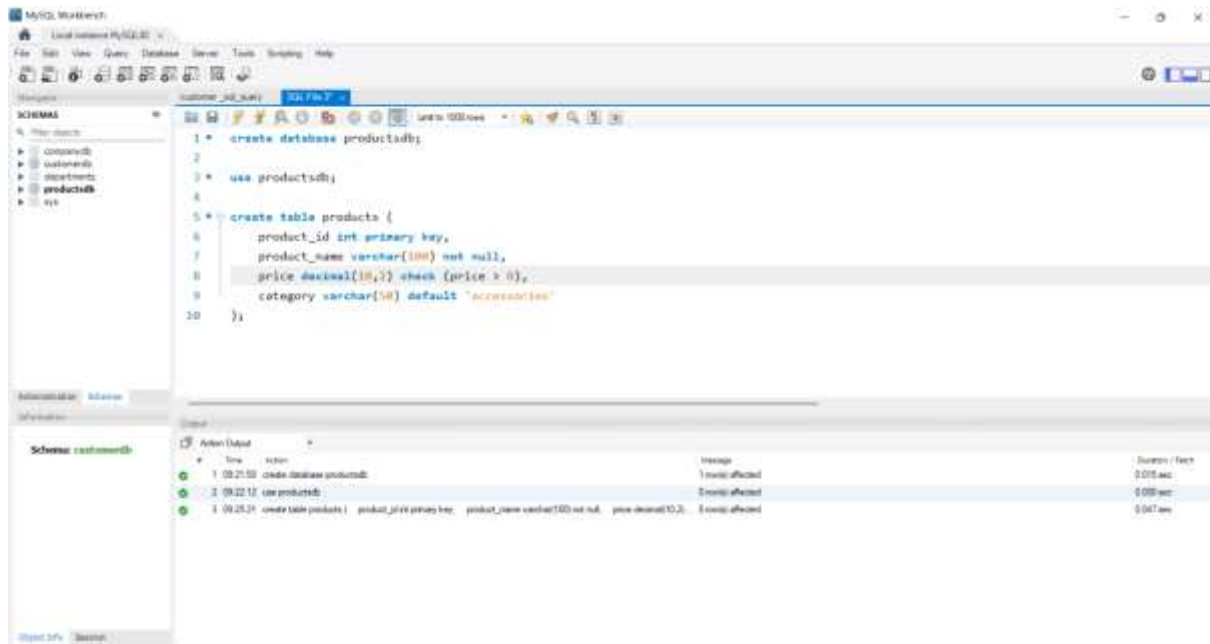
create a products table with

product_id ---->primary key

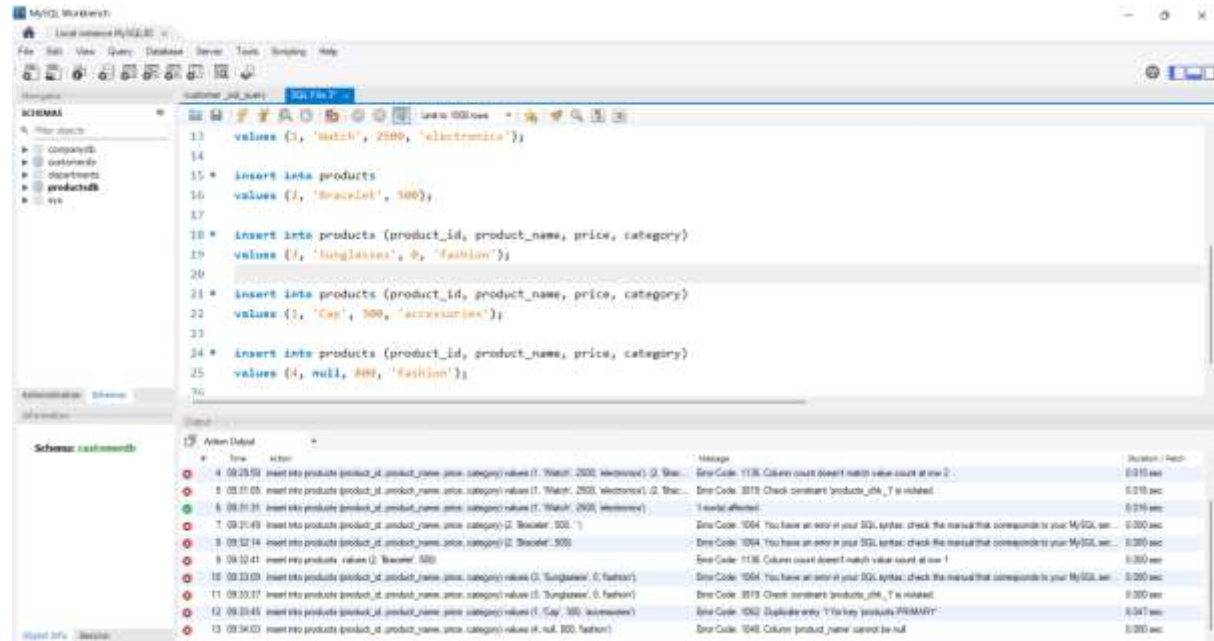
product_name---->not null

price----check>0

category--->default:'accessories'



insert 5 records and test different constraint errors



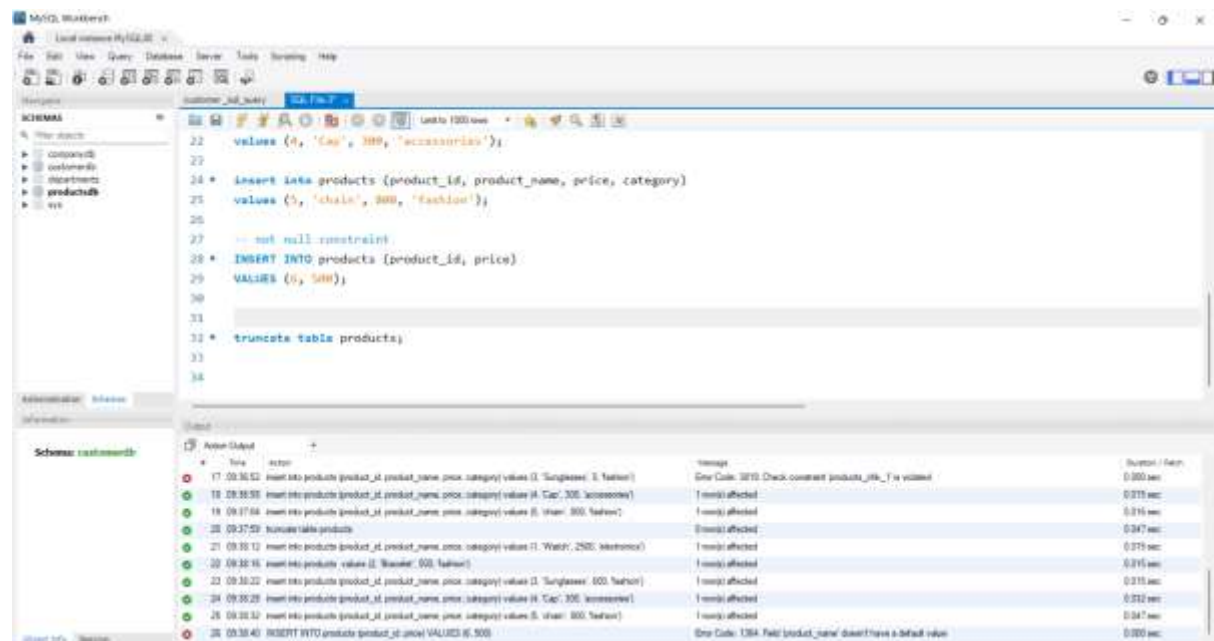
The screenshot shows the MySQL Workbench interface with a SQL editor and a results pane. The SQL editor contains the following queries:

```
13 values (1, 'Watch', 2500, 'electronics');
14
15 * insert into products
16 values (1, 'Bracelet', 500);
17
18 * insert into products (product_id, product_name, price, category)
19 values (1, 'Sunglasses', 0, 'fashion');
20
21 * insert into products (product_id, product_name, price, category)
22 values (1, 'Cap', 500, 'accessories');
23
24 * insert into products (product_id, product_name, price, category)
25 values (4, null, 400, 'fashion');
```

The results pane shows the execution of these queries with the following messages:

Time	Query	Message	Duration / Rows
09:29:59	insert into products (product_id, product_name, price, category) values (1, 'Watch', 2500, 'electronics'); (2) Rows	Error Code: 1138. Column count doesn't match value count at row 2	0.010 sec
09:30:05	insert into products (product_id, product_name, price, category) values (1, 'Watch', 2500, 'electronics'); (2) Rows	Error Code: 3019. Check constraint 'products_pk4_T' is violated	0.016 sec
09:30:36	insert into products (product_id, product_name, price, category) values (1, 'Watch', 2500, 'electronics')	1 row(s) affected	0.016 sec
09:30:49	insert into products (product_id, product_name, price, category) (2) 'Sunglasses', 0, 'fashion'	Error Code: 1064. You have an error in your SQL syntax; check the manual that corresponds to your MySQL ser...	0.000 sec
09:30:54	insert into products (product_id, product_name, price, category) (2) 'Sunglasses', 0, 'fashion'	Error Code: 1064. You have an error in your SQL syntax; check the manual that corresponds to your MySQL ser...	0.000 sec
09:30:54	insert into products values (1, 'Bracelet', 500)	Error Code: 1138. Column count doesn't match value count at row 1	0.000 sec
09:30:59	insert into products (product_id, product_name, price, category) values (1, 'Sunglasses', 0, 'fashion')	Error Code: 1064. You have an error in your SQL syntax; check the manual that corresponds to your MySQL ser...	0.000 sec
09:30:59	insert into products (product_id, product_name, price, category) values (1, 'Sunglasses', 0, 'fashion')	Error Code: 3019. Check constraint 'products_pk4_T' is violated	0.000 sec
09:30:59	insert into products (product_id, product_name, price, category) values (1, 'Cap', 500, 'accessories')	Error Code: 1062. Duplicate entry '1' for key 'products.PRIMARY'	0.047 sec
09:34:03	insert into products (product_id, product_name, price, category) values (4, null, 400, 'fashion')	Error Code: 1048. Column 'product_name' cannot be null	0.000 sec

not null :



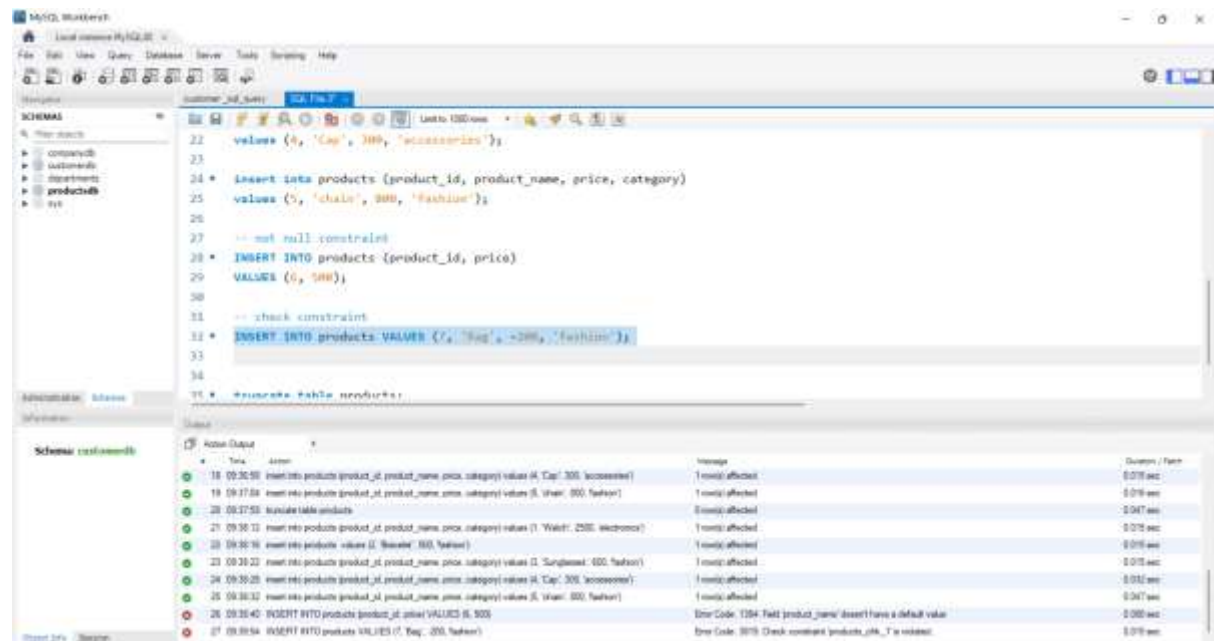
The screenshot shows the MySQL Workbench interface with a SQL editor and a results pane. The SQL editor contains the following queries:

```
22 values (4, 'Cap', 500, 'accessories');
23
24 * insert into products (product_id, product_name, price, category)
25 values (5, 'chain', 500, 'fashion');
26
27 -- not null constraint
28 * INSERT INTO products (product_id, price)
29 VALUES (6, 500);
30
31
32 * truncate table products;
33
34
```

The results pane shows the execution of these queries with the following messages:

Time	Query	Message	Duration / Rows
09:36:52	insert into products (product_id, product_name, price, category) values (1, 'Sunglasses', 0, 'fashion')	Error Code: 3019. Check constraint 'products_pk4_T' is violated	0.000 sec
09:36:56	insert into products (product_id, product_name, price, category) values (4, 'Cap', 500, 'accessories')	1 row(s) affected	0.016 sec
09:37:04	insert into products (product_id, product_name, price, category) values (5, 'chain', 500, 'fashion')	1 row(s) affected	0.016 sec
09:37:29	truncate table products	0 row(s) affected	0.047 sec
09:38:12	insert into products (product_id, product_name, price, category) values (1, 'Watch', 2500, 'electronics')	1 row(s) affected	0.016 sec
09:38:16	insert into products values (1, 'Bracelet', 500, 'fashion')	1 row(s) affected	0.016 sec
09:38:22	insert into products (product_id, product_name, price, category) values (1, 'Sunglasses', 0, 'fashion')	1 row(s) affected	0.016 sec
09:38:28	insert into products (product_id, product_name, price, category) values (4, 'Cap', 500, 'accessories')	1 row(s) affected	0.016 sec
09:38:32	insert into products (product_id, product_name, price, category) values (5, 'chain', 500, 'fashion')	1 row(s) affected	0.047 sec
09:38:40	INSERT INTO products (product_id, price) VALUES (6, 500)	Error Code: 1364. Field 'product_name' doesn't have a default value	0.000 sec

Check constraint:



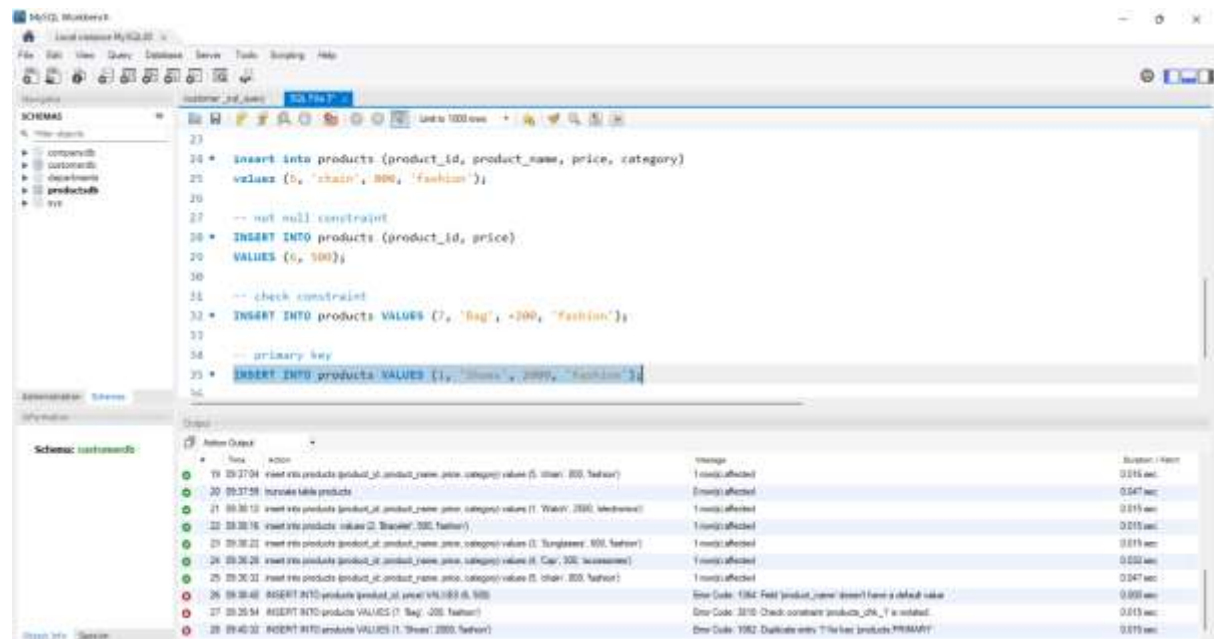
The screenshot shows the MySQL Workbench interface. The SQL editor contains the following queries:

```
22 values (4, 'Cap', 300, 'accessories');
23
24 * insert into products (product_id, product_name, price, category)
25 values (5, 'chain', 800, 'fashion');
26
27 -- not null constraint
28 * INSERT INTO products (product_id, price)
29 VALUES (6, 500);
30
31 -- check constraint
32 * INSERT INTO products VALUES (7, 'Bag', 200, 'Fashion');
```

The Results tab shows the execution output:

Time	Action	Message	Duration / Rows
09:30:58	insert into products (product_id, product_name, price, category) values (4, 'Cap', 300, 'accessories')	1 row(s) affected	0.019 sec
09:31:04	insert into products (product_id, product_name, price, category) values (5, 'chain', 800, 'fashion')	1 row(s) affected	0.019 sec
09:31:05	insert into products (product_id, product_name, price, category) values (6, 500)	1 row(s) affected	0.047 sec
09:31:11	insert into products (product_id, product_name, price, category) values (7, 'Bag', 200, 'Fashion')	1 row(s) affected	0.019 sec
09:31:16	insert into products values (8, 'Shower', 800, 'Fashion')	1 row(s) affected	0.019 sec
09:31:22	insert into products (product_id, product_name, price, category) values (1, 'Sunglass', 600, 'Fashion')	1 row(s) affected	0.019 sec
09:31:28	insert into products (product_id, product_name, price, category) values (3, 'Cap', 300, 'accessories')	1 row(s) affected	0.032 sec
09:31:32	insert into products (product_id, product_name, price, category) values (5, 'chain', 800, 'fashion')	1 row(s) affected	0.047 sec
09:31:40	INSERT INTO products (product_id, price) VALUES (6, 500)	Error Code: 1064 Field 'product_name' doesn't have a default value	0.000 sec
09:31:54	INSERT INTO products VALUES (7, 'Bag', 200, 'Fashion')	Error Code: 8015 Check constraint 'products_chk_1' is violated	0.019 sec

Primary key constraint:



The screenshot shows the MySQL Workbench interface. The SQL editor contains the following queries:

```
23
24 * insert into products (product_id, product_name, price, category)
25 values (5, 'chain', 800, 'fashion');
26
27 -- not null constraint
28 * INSERT INTO products (product_id, price)
29 VALUES (6, 500);
30
31 -- check constraint
32 * INSERT INTO products VALUES (7, 'Bag', 200, 'Fashion');
```

The Results tab shows the execution output:

Time	Action	Message	Duration / Rows
09:31:04	insert into products (product_id, product_name, price, category) values (5, 'chain', 800, 'fashion')	1 row(s) affected	0.019 sec
09:31:05	insert into products (product_id, product_name, price, category) values (6, 500)	1 row(s) affected	0.047 sec
09:31:11	insert into products (product_id, product_name, price, category) values (7, 'Bag', 200, 'Fashion')	1 row(s) affected	0.019 sec
09:31:16	insert into products values (8, 'Shower', 800, 'Fashion')	1 row(s) affected	0.019 sec
09:31:22	insert into products (product_id, product_name, price, category) values (1, 'Sunglass', 600, 'Fashion')	1 row(s) affected	0.019 sec
09:31:28	insert into products (product_id, product_name, price, category) values (3, 'Cap', 300, 'accessories')	1 row(s) affected	0.032 sec
09:31:32	insert into products (product_id, product_name, price, category) values (5, 'chain', 800, 'fashion')	1 row(s) affected	0.047 sec
09:31:40	INSERT INTO products (product_id, price) VALUES (6, 500)	Error Code: 1064 Field 'product_name' doesn't have a default value	0.000 sec
09:31:54	INSERT INTO products VALUES (7, 'Bag', 200, 'Fashion')	Error Code: 1010 Check constraint 'products_chk_1' is violated	0.019 sec
09:40:32	INSERT INTO products VALUES (1, 'Shower', 800, 'Fashion')	Error Code: 1062 Duplicate entry '1' for key 'products.PRIMARY'	0.019 sec

Foreign key:

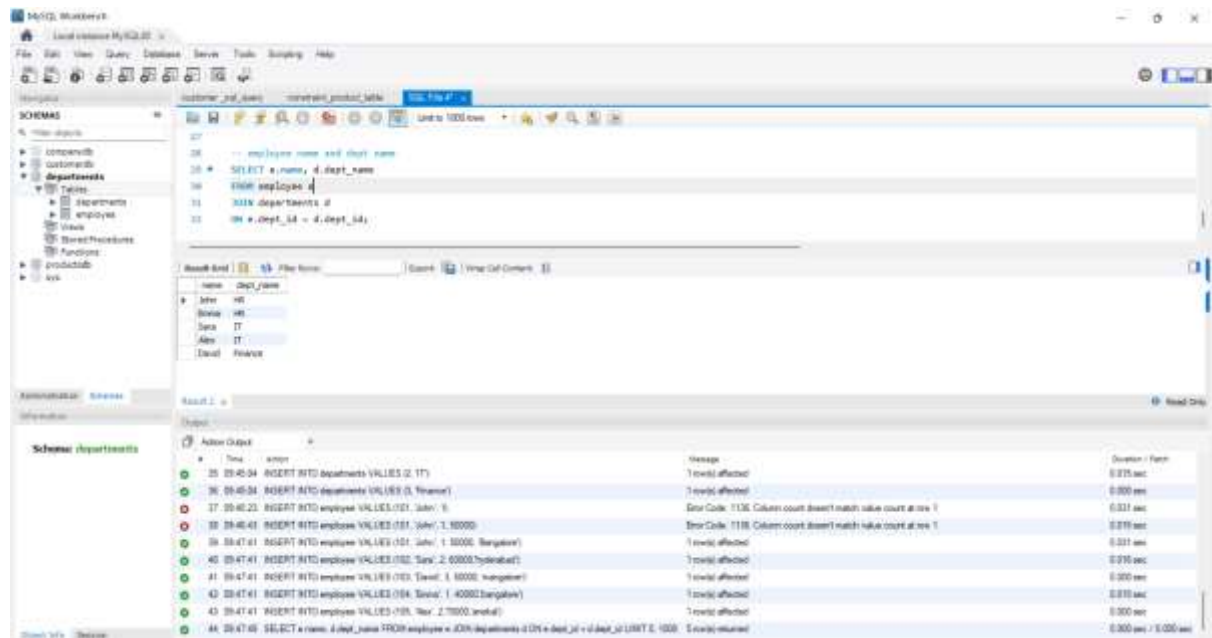
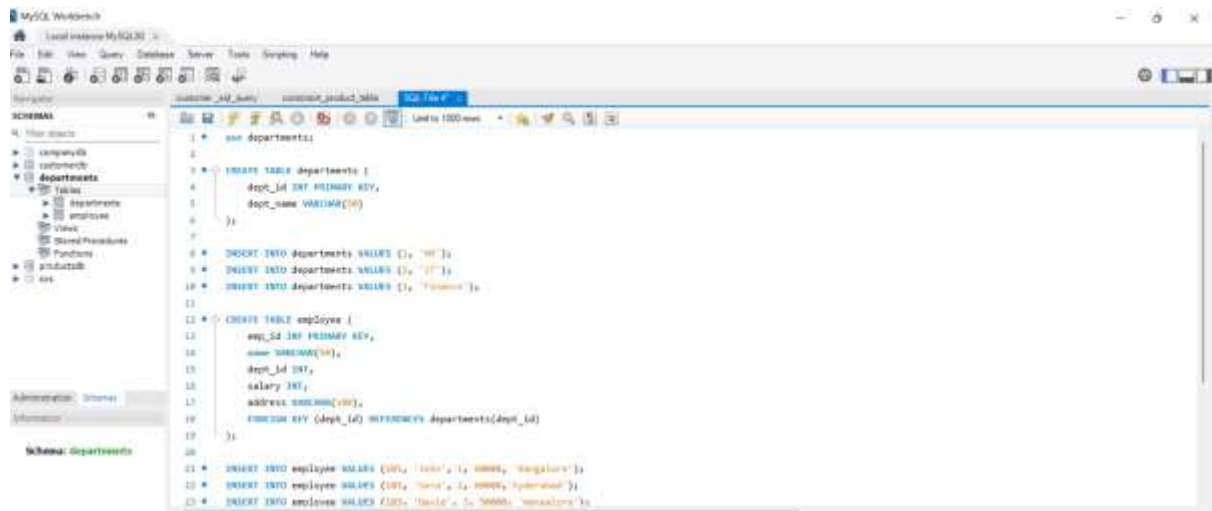
task:

create two tables

employee----->emp_id,name,dept_id,salary,address

departments----->dept_id,dept_name

- 1.find the employee name and department name
- 2.find the employees who don't belong to any department
- 3.find the departments without employees
- 4.find the employee count per department
- 5.find the highest salary in each department
- 6.find the department with more than 2 employees
- 7.find the employees working in the IT department
- 8.find the employees working in the HR department
- 9.find the employees working in the finance department
- 10.find the employees belongs to marketing department
- 11.find the employees belongs to non it department



MySQL Workbench

Local instance MySQL80 v

File Edit View Query Database Server Tools Scripting Help

customer_id_query customer_product_table SQL File v

11 * SELECT e.name, d.dept_name
12 FROM employee e
13 JOIN departments d
14 ON e.dept_id = d.dept_id
15
16 -- employees who don't belong to any department
17 * SELECT name
18 FROM employee
19 WHERE dept_id IS NULL

Result Grid

name

Job

employees 2

Output

Action Output

#	Time	Action	Message	Duration / Peak
42	09:47:41	INSERT INTO employee VALUES (100, 'David', 2, 8000, 'Marketing')	1 row(s) affected	0.016 sec
43	09:47:41	INSERT INTO employee VALUES (103, 'David', 3, 8000, 'Marketing')	1 row(s) affected	0.016 sec
44	09:47:41	INSERT INTO employee VALUES (104, 'David', 1, 4000, 'Marketing')	1 row(s) affected	0.016 sec
45	09:47:41	INSERT INTO employee VALUES (105, 'New', 3, 7000, 'Marketing')	1 row(s) affected	0.016 sec
46	09:47:49	SELECT e.name, d.dept_name FROM employee e JOIN departments d ON e.dept_id = d.dept_id LIMIT 0, 1000	5 row(s) returned	0.002 sec / 0.000 sec
47	09:49:51	INSERT INTO employee VALUES (105, 'New', null, 7000, 'Marketing')	Error Code: 1062 Duplicate entry '105' for key 'PRIMARY'	0.000 sec
48	09:50:13	INSERT INTO employee VALUES (105, 'New', null, 7000, 'Marketing')	Error Code: 1062 Duplicate entry '105' for key 'PRIMARY'	0.000 sec
49	09:50:54	INSERT INTO employee VALUES (105, 'New', null, 7000, 'Marketing')	Error Code: 1062 Duplicate entry '105' for key 'PRIMARY'	0.000 sec
50	09:51:10	INSERT INTO employee VALUES (105, 'New', null, 7000, 'Marketing')	1 row(s) affected	0.016 sec
51	09:51:52	SELECT name FROM employee WHERE dept_id IS NULL LIMIT 0, 1000	1 row(s) returned	0.000 sec / 0.000 sec

Schema: departments

MySQL Workbench

Local instance MySQL80 v

File Edit View Query Database Server Tools Scripting Help

customer_id_query customer_product_table SQL File v

11 FROM employee
12 WHERE dept_id IS NULL
13
14 -- departments without employees
15 * SELECT d.dept_name
16 FROM departments d
17 LEFT JOIN employee e
18 ON e.dept_id = d.dept_id
19 WHERE e.dept_id IS NULL

Result Grid

dept_name

Marketing

employees 2

Output

Action Output

#	Time	Action	Message	Duration / Peak
42	09:47:41	INSERT INTO employee VALUES (104, 'David', 1, 4000, 'Marketing')	1 row(s) affected	0.016 sec
43	09:47:41	INSERT INTO employee VALUES (105, 'New', 3, 7000, 'Marketing')	1 row(s) affected	0.016 sec
44	09:47:49	SELECT e.name, d.dept_name FROM employee e JOIN departments d ON e.dept_id = d.dept_id LIMIT 0, 1000	5 row(s) returned	0.002 sec / 0.000 sec
45	09:49:51	INSERT INTO employee VALUES (105, 'New', null, 7000, 'Marketing')	Error Code: 1062 Duplicate entry '105' for key 'PRIMARY'	0.000 sec
46	09:50:13	INSERT INTO employee VALUES (105, 'New', null, 7000, 'Marketing')	Error Code: 1062 Duplicate entry '105' for key 'PRIMARY'	0.000 sec
47	09:50:54	INSERT INTO employee VALUES (105, 'New', null, 7000, 'Marketing')	Error Code: 1062 Duplicate entry '105' for key 'PRIMARY'	0.000 sec
48	09:51:10	INSERT INTO employee VALUES (105, 'New', null, 7000, 'Marketing')	1 row(s) affected	0.016 sec
49	09:51:52	SELECT name FROM employee WHERE dept_id IS NULL LIMIT 0, 1000	1 row(s) returned	0.000 sec / 0.000 sec
50	09:52:01	INSERT INTO departments VALUES (4, 'Marketing')	1 row(s) affected	0.016 sec
51	09:52:58	SELECT d.dept_name FROM departments d LEFT JOIN employee e ON e.dept_id = d.dept_id WHERE e.dept_id IS NULL	1 row(s) returned	0.000 sec / 0.000 sec

Schema: departments

MySQL Workbench

Local instance MySQL8.0.32

File Edit View Query Database Server Tools Window Help

customer_dept_name constrain_product_table SQL File P

```

41 -- dept_name = emp_dept TO NULL
42
43 -- employee count per department
44 * SELECT d.dept_name, COUNT(e.emp_id) AS employee_count
45 FROM departments d
46 LEFT JOIN employees e
47 ON d.dept_id = e.dept_id
48 GROUP BY d.dept_name

```

Result grid

dept_name	employee_count
HR	7
IT	3
Finance	1
Marketing	0

Result 1: 0

Output

Action Output

Step	Action	Message	Duration / Rows
41	41: 09:41:41: INSERT INTO employees (e.emp_id, e.dept_name, e.job_id, e.salary, e.commission_pct, e.last_name, e.first_name, e.email, e.phone_number, e.hire_date, e.job_id, e.dept_id) VALUES (100, 'HR', 'J', '10000', NULL, 'J', 'J', 'JJ', '1000', '2000-01-01', 'J', '10000')	1 row(s) affected	0:00 sec / 1:000 sec
44	44: 09:41:45: SELECT e.name, d.dept_name FROM employees e JOIN departments d ON e.dept_id = d.dept_id LIMIT 5: 1000	5 row(s) returned	0:00 sec / 0:000 sec
45	45: 09:45:50: INSERT INTO employees VALUES (105, 'HR', 'M', '70000', NULL, 'M', 'M', 'MM', '70000', '2000-01-01', 'M', '70000')	Error Code: 1062. Duplicate entry '105' for key 'employees.PRIMARY'	0:00 sec
46	46: 09:50:13: INSERT INTO employees VALUES (106, 'HR', 'A', '70000', NULL, 'A', 'A', 'AA', '70000', '2000-01-01', 'A', '70000')	Error Code: 1062. Duplicate entry '106' for key 'employees.PRIMARY'	0:00 sec
47	47: 09:50:54: INSERT INTO employees (e.emp_id, e.name, e.salary, e.address) VALUES (108, 'HR', '70000', NULL, '70000', '2000-01-01', '70000', '2000-01-01')	Error Code: 1062. Duplicate entry '108' for key 'employees.PRIMARY'	0:00 sec
49	49: 09:51:10: INSERT INTO employees (e.emp_id, e.name, e.salary, e.address) VALUES (109, 'HR', '70000', NULL, '70000', '2000-01-01', '70000', '2000-01-01')	1 row(s) affected	0:01 sec
49	49: 09:51:52: SELECT name FROM employee WHERE dept_id IS NULL LIMIT 5: 1000	1 row(s) returned	0:00 sec / 0:000 sec
50	50: 09:52:51: INSERT INTO departments VALUES (4, 'Marketing')	1 row(s) affected	0:01 sec
51	51: 09:52:56: SELECT d.dept_name FROM departments d LEFT JOIN employees e ON d.dept_id = e.dept_id WHERE e emp	1 row(s) returned	0:00 sec / 0:000 sec
52	52: 09:53:58: SELECT d.dept_name, COUNT(e.emp_id) AS employee_count FROM departments d LEFT JOIN employees e	4 row(s) returned	0:01 sec / 0:000 sec

MySQL Workbench

Local instance MySQL8.0.32

File Edit View Query Database Server Tools Window Help

customer_dept_name constrain_product_table SQL File P

```

54 GROUP BY d.dept_name
55
56 -- Highest salary in each department
57 * SELECT d.dept_name, MAX(e.salary) AS highest_salary
58 FROM employees e
59 LEFT JOIN departments d
60 ON e.dept_id = d.dept_id
61 GROUP BY d.dept_name

```

Result grid

dept_name	highest_salary
HR	90000
IT	90000
Finance	90000

Result 1: 0

Output

Action Output

Step	Action	Message	Duration / Rows
44	44: 09:41:45: SELECT e.name, d.dept_name FROM employees e JOIN departments d ON e.dept_id = d.dept_id LIMIT 5: 1000	5 row(s) returned	0:00 sec / 0:000 sec
45	45: 09:45:50: INSERT INTO employees VALUES (105, 'HR', 'M', '70000', NULL, 'M', 'M', 'MM', '70000', '2000-01-01', 'M', '70000')	Error Code: 1062. Duplicate entry '105' for key 'employees.PRIMARY'	0:00 sec
46	46: 09:50:13: INSERT INTO employees VALUES (106, 'HR', 'A', '70000', NULL, 'A', 'A', 'AA', '70000', '2000-01-01', 'A', '70000')	Error Code: 1062. Duplicate entry '106' for key 'employees.PRIMARY'	0:00 sec
47	47: 09:50:54: INSERT INTO employees (e.emp_id, e.name, e.salary, e.address) VALUES (108, 'HR', '70000', NULL, '70000', '2000-01-01', '70000', '2000-01-01')	Error Code: 1062. Duplicate entry '108' for key 'employees.PRIMARY'	0:00 sec
49	49: 09:51:10: INSERT INTO employees (e.emp_id, e.name, e.salary, e.address) VALUES (109, 'HR', '70000', NULL, '70000', '2000-01-01', '70000', '2000-01-01')	1 row(s) affected	0:01 sec
49	49: 09:51:52: SELECT name FROM employee WHERE dept_id IS NULL LIMIT 5: 1000	1 row(s) returned	0:00 sec / 0:000 sec
50	50: 09:52:51: INSERT INTO departments VALUES (4, 'Marketing')	1 row(s) affected	0:01 sec
51	51: 09:52:56: SELECT d.dept_name FROM departments d LEFT JOIN employees e ON d.dept_id = e.dept_id WHERE e emp	1 row(s) returned	0:00 sec / 0:000 sec
52	52: 09:53:58: SELECT d.dept_name, COUNT(e.emp_id) AS employee_count FROM departments d LEFT JOIN employees e	4 row(s) returned	0:01 sec / 0:000 sec
53	53: 09:54:36: SELECT d.dept_name, MAX(e.salary) AS highest_salary FROM employees e JOIN departments d ON e.dept_id = d.dept_id	3 row(s) returned	0:00 sec / 0:000 sec

MySQL Workbench

Local instance MySQL8.0.33

File Edit View Query Database Server Tools Sampling Help

customer_dept_name constrain_product_table SQL Tool F

Limit to 1000 rows

```

42 -- department with more than 2 employees
43 SELECT d.dept_name
44 FROM employee e
45 JOIN departments d
46 ON e.dept_id = d.dept_id
47 GROUP BY d.dept_name
48 HAVING COUNT(e.emp_id) > 2

```

Result grid

dept_name
IT

Schema: departments

Information

Schema: departments

#	Time	Action	Message	Duration / Rows
46	00:00:13	INSERT INTO employee VALUES (105, 'Bob', 47000, 'analyst')	Error Code: 1062 Duplicate entry '105' for key 'employee.PRIMARY'	0.000 sec
47	00:00:14	INSERT INTO employee VALUES (106, 'Bob', 70000, 'analyst')	Error Code: 1062 Duplicate entry '106' for key 'employee.PRIMARY'	0.000 sec
48	00:01:10	INSERT INTO employee VALUES (107, 'Bob', 70000, 'analyst')	1 row(s) affected	0.015 sec
49	00:01:52	SELECT name FROM employee WHERE dept_id IS NULL LIMIT 5 1000	1 row(s) returned	0.000 sec / 0.000 sec
50	00:02:51	INSERT INTO departments VALUES (4, 'Marketing')	1 row(s) affected	0.016 sec
51	00:02:58	SELECT d.dept_name FROM departments d LEFT JOIN employee e ON d.dept_id = e.dept_id WHERE e.emp_id IS NULL	1 row(s) returned	0.000 sec / 0.000 sec
52	00:03:58	SELECT d.dept_name, COUNT(e.emp_id) AS employee_count FROM departments d LEFT JOIN employee e ON d.dept_id = e.dept_id GROUP BY d.dept_name	4 row(s) returned	0.016 sec / 0.000 sec
53	00:04:36	SELECT d.dept_name, MAX(e.salary) AS highest_salary FROM employee e JOIN departments d ON e.dept_id = d.dept_id GROUP BY d.dept_name	3 row(s) returned	0.000 sec / 0.000 sec
54	00:06:06	INSERT INTO employee VALUES (107, 'New', 27000, 'analyst')	1 row(s) affected	0.010 sec
55	00:06:12	SELECT d.dept_name FROM employee e JOIN departments d ON e.dept_id = d.dept_id GROUP BY d.dept_name	1 row(s) returned	0.000 sec / 0.000 sec

MySQL Workbench

Local instance MySQL8.0.33

File Edit View Query Database Server Tools Sampling Help

customer_dept_name constrain_product_table SQL Tool F

Limit to 1000 rows

```

49 -- employees working in the IT department
50 SELECT e.name
51 FROM employee e
52 JOIN departments d
53 ON e.dept_id = d.dept_id
54 WHERE d.dept_name = 'IT'

```

Result grid

name
Bob
Ann

Schema: departments

Information

Schema: departments

#	Time	Action	Message	Duration / Rows
47	00:00:14	INSERT INTO employee VALUES (105, 'Bob', 70000, 'analyst')	Error Code: 1062 Duplicate entry '105' for key 'employee.PRIMARY'	0.000 sec
48	00:01:10	INSERT INTO employee VALUES (106, 'Bob', 70000, 'analyst')	1 row(s) affected	0.015 sec
49	00:01:52	SELECT name FROM employee WHERE dept_id IS NULL LIMIT 5 1000	1 row(s) returned	0.000 sec / 0.000 sec
50	00:02:51	INSERT INTO departments VALUES (4, 'Marketing')	1 row(s) affected	0.016 sec
51	00:02:58	SELECT d.dept_name FROM departments d LEFT JOIN employee e ON d.dept_id = e.dept_id WHERE e.emp_id IS NULL	1 row(s) returned	0.000 sec / 0.000 sec
52	00:03:58	SELECT d.dept_name, COUNT(e.emp_id) AS employee_count FROM departments d LEFT JOIN employee e ON d.dept_id = e.dept_id GROUP BY d.dept_name	4 row(s) returned	0.016 sec / 0.000 sec
53	00:04:36	SELECT d.dept_name, MAX(e.salary) AS highest_salary FROM employee e JOIN departments d ON e.dept_id = d.dept_id GROUP BY d.dept_name	3 row(s) returned	0.000 sec / 0.000 sec
54	00:06:06	INSERT INTO employee VALUES (107, 'New', 27000, 'analyst')	1 row(s) affected	0.010 sec
55	00:06:12	SELECT d.dept_name FROM employee e JOIN departments d ON e.dept_id = d.dept_id GROUP BY d.dept_name	1 row(s) returned	0.000 sec / 0.000 sec
56	00:06:51	SELECT e.name FROM employee e JOIN departments d ON e.dept_id = d.dept_id WHERE d.dept_name = 'IT'	2 row(s) returned	0.000 sec / 0.000 sec

MySQL Workbench

Local instance MySQL8.0.33

File Edit View Query Database Server Tools Window Help

customer_jst_sales constraint_product_table SQL File P

Limit to 1000 rows

```

76 WHERE d.dept_name = 'IT';
77
78 -- employees working in the HR department
79 *
80 SELECT e.name
81 FROM employee e
82 JOIN departments d
83 ON e.dept_id = d.dept_id
84 WHERE d.dept_name = 'HR';

```

Result grid

name
John

Result 1

Output

Action Output

#	Msg	Message	Duration / Secs
46	09:51:10: INSERT INTO employee (name, salary, address) VALUES ('SC', 70000, 'null');	1 row(s) affected	0.016 sec
47	09:51:52: SELECT name FROM employee WHERE dept_id IS NULL LIMIT 5, 1000;	1 row(s) returned	0.000 sec / 0.000 sec
50	09:52:51: INSERT INTO departments (VALUES ('Marketing'));	1 row(s) affected	0.016 sec
51	09:52:52: SELECT d.dept_name FROM departments d LEFT JOIN employee e ON d.dept_id = e.dept_id WHERE e emp...	1 row(s) returned	0.000 sec / 0.000 sec
52	09:52:53: SELECT d.dept_name, COUNT(*) emp_cnt AS employee_count FROM departments d LEFT JOIN employee e...	4 row(s) returned	0.016 sec / 0.000 sec
53	09:54:36: SELECT d.dept_name, MAX(s.salary) AS highest_salary FROM employee e JOIN departments d ON e.dept_id...	3 row(s) returned	0.000 sec / 0.000 sec
54	09:56:06: INSERT INTO employee VALUES ('ST', 'Nail', 27000, 'null');	1 row(s) affected	0.012 sec
55	09:56:12: SELECT d.dept_name FROM employee e JOIN departments d ON e.dept_id = d.dept_id GROUP BY d.dept_n...	1 row(s) returned	0.000 sec / 0.000 sec
56	09:56:51: SELECT e.name FROM employee e JOIN departments d ON e.dept_id = d.dept_id WHERE d.dept_name = 'IT...	3 row(s) returned	0.000 sec / 0.000 sec
57	09:57:31: SELECT e.name FROM employee e JOIN departments d ON e.dept_id = d.dept_id WHERE d.dept_name = 'H...	2 row(s) returned	0.000 sec / 0.000 sec
58	09:57:31: SELECT e.name FROM employee e JOIN departments d ON e.dept_id = d.dept_id WHERE d.dept_name = 'H...	2 row(s) returned	0.000 sec / 0.000 sec

Schema: departments

Tables: departments, employee, views, Stored Procedures, Functions, products, etc.

MySQL Workbench

Local instance MySQL8.0.33

File Edit View Query Database Server Tools Window Help

customer_jst_sales constraint_product_table SQL File P

Limit to 1000 rows

```

87 WHERE d.dept_name = 'HR';
88
89 -- employees working in the Finance department
90 *
91 SELECT e.name
92 FROM employee e
93 JOIN departments d
94 ON e.dept_id = d.dept_id
95 WHERE d.dept_name = 'Finance';

```

Result grid

name
David

Result 12

Output

Action Output

#	Msg	Message	Duration / Secs
49	09:51:52: SELECT name FROM employee WHERE dept_id IS NULL LIMIT 5, 1000;	1 row(s) returned	0.000 sec / 0.000 sec
50	09:52:51: INSERT INTO departments (VALUES ('Marketing'));	1 row(s) affected	0.016 sec
51	09:52:52: SELECT d.dept_name FROM departments d LEFT JOIN employee e ON d.dept_id = e.dept_id WHERE e emp...	1 row(s) returned	0.000 sec / 0.000 sec
52	09:52:53: SELECT d.dept_name, COUNT(*) emp_cnt AS employee_count FROM departments d LEFT JOIN employee e...	4 row(s) returned	0.016 sec / 0.000 sec
53	09:54:36: SELECT d.dept_name, MAX(s.salary) AS highest_salary FROM employee e JOIN departments d ON e.dept_id...	3 row(s) returned	0.000 sec / 0.000 sec
54	09:56:06: INSERT INTO employee VALUES ('ST', 'Nail', 27000, 'null');	1 row(s) affected	0.012 sec
55	09:56:12: SELECT d.dept_name FROM employee e JOIN departments d ON e.dept_id = d.dept_id GROUP BY d.dept_n...	1 row(s) returned	0.000 sec / 0.000 sec
56	09:56:51: SELECT e.name FROM employee e JOIN departments d ON e.dept_id = d.dept_id WHERE d.dept_name = 'IT...	3 row(s) returned	0.000 sec / 0.000 sec
57	09:57:31: SELECT e.name FROM employee e JOIN departments d ON e.dept_id = d.dept_id WHERE d.dept_name = 'H...	2 row(s) returned	0.000 sec / 0.000 sec
58	09:58:06: SELECT e.name FROM employee e JOIN departments d ON e.dept_id = d.dept_id WHERE d.dept_name = 'H...	1 row(s) returned	0.000 sec / 0.000 sec

Schema: departments

Tables: departments, employee, views, Stored Procedures, Functions, products, etc.

MySQL Workbench

Local instance MySQL80

File Edit View Query Database Server Tools Window Help

customer_dept_admin constraint_product_table SQL File P

Limit to 1000 rows

```

50  WHERE d.dept_name = 'Marketing'
51
52  -- employees belonging to Marketing department
53  *
54  SELECT e.name
55  FROM employee e
56  JOIN departments d
57  ON e.dept_id = d.dept_id
58  WHERE d.dept_name = 'Marketing'

```

Result grid

name

Result 12

Output

Action Output

#	Time	Message	Duration / Rows
50	00:00:51	INSERT INTO departments VALUES (4, 'Marketing')	1 row(s) affected
51	00:00:50	SELECT d.dept_name FROM departments d LEFT JOIN employee e ON d.dept_id = e.dept_id WHERE e.emp...	1 row(s) returned
52	00:00:50	SELECT d.dept_name, COUNT(e.emp_id) AS employee_count FROM departments d LEFT JOIN employee e...	4 row(s) returned
53	00:00:50	SELECT d.dept_name, MAX(e.salary) AS highest_salary FROM employee e JOIN departments d ON e.dept_id =...	3 row(s) returned
54	00:00:50	INSERT INTO employee VALUES (107, 'New', 17000, 'new')	1 row(s) affected
55	00:00:52	SELECT d.dept_name FROM employee e JOIN departments d ON e.dept_id = d.dept_id GROUP BY d.dept_n...	1 row(s) returned
56	00:00:51	SELECT e.name FROM employee e JOIN departments d ON e.dept_id = d.dept_id WHERE d.dept_name = 'T...	2 row(s) returned
57	00:00:51	SELECT e.name FROM employee e JOIN departments d ON e.dept_id = d.dept_id WHERE d.dept_name = 'T...	2 row(s) returned
58	00:00:50	SELECT e.name FROM employee e JOIN departments d ON e.dept_id = d.dept_id WHERE d.dept_name = 'M...	2 row(s) returned
59	00:00:51	SELECT e.name FROM employee e JOIN departments d ON e.dept_id = d.dept_id WHERE d.dept_name = 'M...	2 row(s) returned
60	00:00:51	SELECT e.name FROM employee e JOIN departments d ON e.dept_id = d.dept_id WHERE d.dept_name = 'M...	2 row(s) returned

Schema: departments

MySQL Workbench

Local instance MySQL80

File Edit View Query Database Server Tools Window Help

customer_dept_admin constraint_product_table SQL File P

Limit to 1000 rows

```

57  WHERE d.dept_name = 'Marketing'
58
59  -- employees belonging to new IT department
60  *
61  SELECT e.name
62  FROM employee e
63  JOIN departments d
64  ON e.dept_id = d.dept_id
65  WHERE d.dept_name = 'IT'

```

Result grid

name

John
Diana
David

Result 12

Output

Action Output

#	Time	Message	Duration / Rows
51	00:00:50	SELECT d.dept_name FROM departments d LEFT JOIN employee e ON d.dept_id = e.dept_id WHERE e.emp...	1 row(s) returned
52	00:00:50	SELECT d.dept_name, COUNT(e.emp_id) AS employee_count FROM departments d LEFT JOIN employee e...	4 row(s) returned
53	00:00:50	SELECT d.dept_name, MAX(e.salary) AS highest_salary FROM employee e JOIN departments d ON e.dept_id =...	3 row(s) returned
54	00:00:50	INSERT INTO employee VALUES (107, 'New', 17000, 'new')	1 row(s) affected
55	00:00:52	SELECT d.dept_name FROM employee e JOIN departments d ON e.dept_id = d.dept_id GROUP BY d.dept_n...	1 row(s) returned
56	00:00:51	SELECT e.name FROM employee e JOIN departments d ON e.dept_id = d.dept_id WHERE d.dept_name = 'T...	2 row(s) returned
57	00:00:51	SELECT e.name FROM employee e JOIN departments d ON e.dept_id = d.dept_id WHERE d.dept_name = 'T...	2 row(s) returned
58	00:00:50	SELECT e.name FROM employee e JOIN departments d ON e.dept_id = d.dept_id WHERE d.dept_name = 'M...	2 row(s) returned
59	00:00:51	SELECT e.name FROM employee e JOIN departments d ON e.dept_id = d.dept_id WHERE d.dept_name = 'M...	2 row(s) returned
60	00:00:50	SELECT e.name FROM employee e JOIN departments d ON e.dept_id = d.dept_id WHERE d.dept_name = 'I...	3 row(s) returned

Schema: departments